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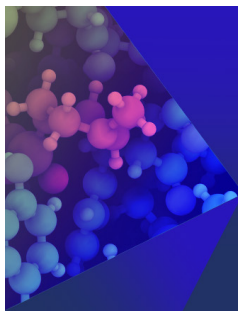
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Glassy dynamics in mutant huntingtin proteins

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Causative to the neurodegenerative Huntington's disease (HD), a mutational huntingtin (HTT) protein consists of an unusual expansion on the poly-glutamine (polyQ) region in the first exon (exon-1) domain. Despite its significance on HD progression, the structural role of the exon-1 with the polyQ region is still elusive. As HTT is an intrinsically disordered protein (IDP), a large ensemble of various conformations (instead of a mostly single native conformation) is required to characterize its structural properties and to infer biological functions, which is challenging even for the most state-of-the-art experimental techniques. For this reason, molecular dynamics (MD) simulations with enhanced sampling techniques are ideal to complement experiment on collecting such a large ensemble of thermodynamically accessible structures. Here, we performed large-scale temperature replica-exchange MD (T-REMD) simulations on the exon-1 with an illustration on the necessity of using T-REMD instead of unbiased regular MD. By comparing T-REMD data and unbiased MD data, we discovered that (1) the dynamics of polyQ regions are extremely sluggish and glassy at the room temperature and the relaxation of the system cannot be achieved within a reasonable amount of time without utilizing an enhanced sampling method and (2) an ensemble of protein structures containing the surprising *cis*-peptide bonds in the proline-rich domain can be obtained at much elevated temperatures. Our results may provide valuable insights for future studies on the HTT as well as other IDPs using the T-REMD method. *Published by AIP Publishing.* <https://doi.org/10.1063/1.5029369>

INTRODUCTION

A huntingtin (HTT) gene/protein is well known to play key roles in the neurodegenerative Huntington's Disease (HD)¹ that is devastating, lethal, and incurable so far. A mutational expansion of the poly-glutamine (polyQ) region in the HTT triggers the apoptosis of neuronal cells resulting in degeneration and even death of neurons.^{1,2} Although the genetic correlation between the HTT and the occurrence of HD has been elucidated in several gene-association studies,^{3–5} it is still elusive how the HTT with an expanded polyQ region can cause the HD.

Several experimental studies have been devoted to identifying the structural motifs relevant to HD pathogens. Especially, the structure of the first exon (exon-1) of the HTT (httex1) containing an expanded polyQ region has gained much attention^{6–11} because of its direct genetic and proteomic association with the HD.^{12–15} Nuclear magnetic resonance (NMR) studies showed that the first 17 residues (N17) exist in alpha helical forms when it contacts with micelles⁷ or maltose binding proteins (MBP),¹¹ but it remains disordered in solution.^{16,17} Numerical studies also confirmed these findings in both micelle and solution environments.¹⁸ By contrast, a proline-rich region consisting of 38 residues (C38) followed by polyQ remains in rigid Poly-proline II (PPII) helices connected by a hinge region most of the time.¹⁹

On the other hand, the polyQ structure has not been fully identified in experiment yet. For instance, it was reported that the unmodified httex1 diffracted at a poor resolution, implying the structural disorder in the polyQ region to a certain extent.⁹ NMR and cryo-Electron Microscopy (cryo-EM) studies also showed that the polyQ chains exist in coil or unstructured forms.^{10,20} These experimental observations strongly suggest that the polyQ region does not have a single native state but rather explores many different conformations. Such an intrinsic disorder poses difficulties in studying the structure and function of the httex1, as we need to collect the extensive ensemble of various conformations of the httex1.^{21,22} Molecular dynamics (MD) simulations with enhanced sampling techniques may play an important role here by facilitating reasonable equilibration, improving the quality of sampling, and enabling the observation of many meta-stable conformations.

Among various enhanced sampling techniques, temperature replica exchange molecular dynamics (T-REMD) is one of the most powerful methods.²³ T-REMD starts with multiple numbers of replicas set up at different temperatures. By random-walking through various temperatures, each replica can reach from room-to high-temperatures letting the system overcome thermodynamic barriers and escape from local minima. Indeed, T-REMD has been proven to be a straightforward enhanced sampling method that is theoretically rigorous in the study of protein structures,^{24–28} however, it also has several limitations. For instance, T-REMD requires a relatively large number of replicas in order to have reasonable acceptance ratios for a relatively wide temperature range. Another related enhanced sampling method is the Replica Exchange

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with Solute Tempering (REST),²⁹ where the solution remains at room temperature while the solute is heated, so it requires much less replicas; however, for the REST method, only the data at room temperature (solution temperature) are thermodynamically meaningful while all others are basically artificial. Therefore, T-REMD is still preferred for obtaining thermodynamic data, particularly for cases where simulation data at other temperatures are needed.

In this paper, we performed all-atom T-REMD simulations, with 128 replicas, to sample various possible structures of the mutant httex1 with 56 consecutive glutamines (Q56). Simulated temperatures ranging from 298 K to 600 K allow us to obtain proper thermodynamic ensembles. Through various analyses, we demonstrate that T-REMD simulations are not only helpful but also essential for the study of httex1 and its polyQ tracts. Especially, our results manifest that because of the extremely sluggish dynamics of polyQ tracts, it is not feasible to obtain appropriate the thermodynamics ensemble of httex1 structures without enhanced sampling methods.

METHODS

The random initial structures of the human httex1 Q56 (sequence: MATLEKLMKAFESLKSF-Q56-P11QLPQPPPQAQPLLQPQP10) are prepared as follows. We generated the fully extended conformations of the httex1. Then, the protein was collapsed in the implicit solvent model (generalized Born/surface area method) and was thereafter solvated with TIP3P water.³⁰ The dimension of the simulation box was defined with a 2 nm margin from the largest protein conformation. The system is subsequently placed at 600 K for more than 150 ns. The initial structures for 128 replicas were taken from the long 600 K simulation data, about one structure every 1 ns. T-REMD with 128 replicas was performed with the standard procedure.²³ Acceptance ratio was approximately 0.3. Temperature ranges from 298 K to about 600 K. The simulation was done under the NVT condition, and the size of the simulation box was $81.72 \times 81.72 \times 81.72 \text{ \AA}^3$. The Optimized Potentials for Liquid Simulations All-Atom (OPLS/AA) force field was used for httex1.³¹ All simulations were performed with GROMINGEN MACHINE for Chemical Simulations (GROMACS) 5.0.5 simulation package.³² Detailed simulation protocols can be found in our previous work.¹⁹

In addition to T-REMD simulations, we performed separate unbiased MD simulations to investigate the dynamics. The unbiased MD simulations were started from representative structures taken from the T-REMD Q56 trajectory. The representative structures were selected from the most populated clusters classified by the GROMINGEN MOLECULAR SIMULATION (GROMOS) algorithm³³ and with the root-mean-squared distances less than 3.5 Å. We used the same force field and simulation setup parameters for the unbiased MD as those used in T-REMD.

RESULTS

Convergence test

The general simulation setup is described in Fig. 1. We ensured the convergence of our simulation results using

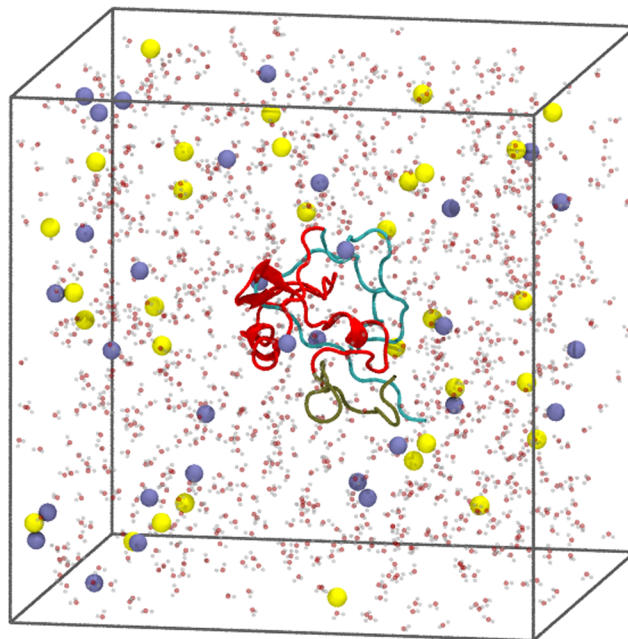


FIG. 1. General simulation setup. Huntingtin exon-1 (httex1) is presented in new cartoon representation in the middle. The first 17 residues (N17), poly-glutamine (polyQ), and proline-rich domain (C38) are colored in tan, red, and cyan, respectively. The water is drawn in van der Waals representation with 90% scaling for clarity. Sodium and chloride ions are shown in blue and yellow spheres, respectively.

multiple criteria. First, we investigated the acceptance ratio of simulation results. Our aimed acceptance rate was about 0.2–0.3 for all replicas, which was proven to be the optimal acceptance ratios.^{25,26,28,34} Figure 2(a) confirms that all acceptance ratios appear in this targeted range. Second, the temperature “trace” of each replica was examined. To achieve the best convergences, all replicas should in principle visit all simulated temperatures rather than remain in a small range of temperatures. For each of the three randomly selected replicas, Fig. 2(b) shows that the respective temperature traces indeed span the entire range, indicating a potentially good convergence.

Another measure of the mixing quality is the replica mixing parameter, $m(T)$, developed by Han and Hansmann.³⁵ The mixing parameter, $m(T)$, is defined as

$$m(T) = 1 - \frac{\sqrt{\langle t^2 \rangle / N_R}}{\langle t \rangle},$$

where $\langle t \rangle$ is the average over replicas, t is the time each replica spends at temperature T , and N_R is the total number of replicas in the simulations. In our case, $N_R = 128$. If all replicas spend an equal amount of time at each temperature, $m(T) = 1 - N_R^{-1/2}$. Figure 2(c) shows that $m(T)$ is close to the theoretical prediction, $1 - N_R^{-1/2} = 0.9116$ (red dashed line), over all temperatures we simulated. These data indicate that our temperature sequence presented in Table S1 of the supplementary material generates reasonable acceptance ratios with excellent mixing efficiency.

Finally, we checked the convergence of secondary structures by analyzing secondary structure collective variables

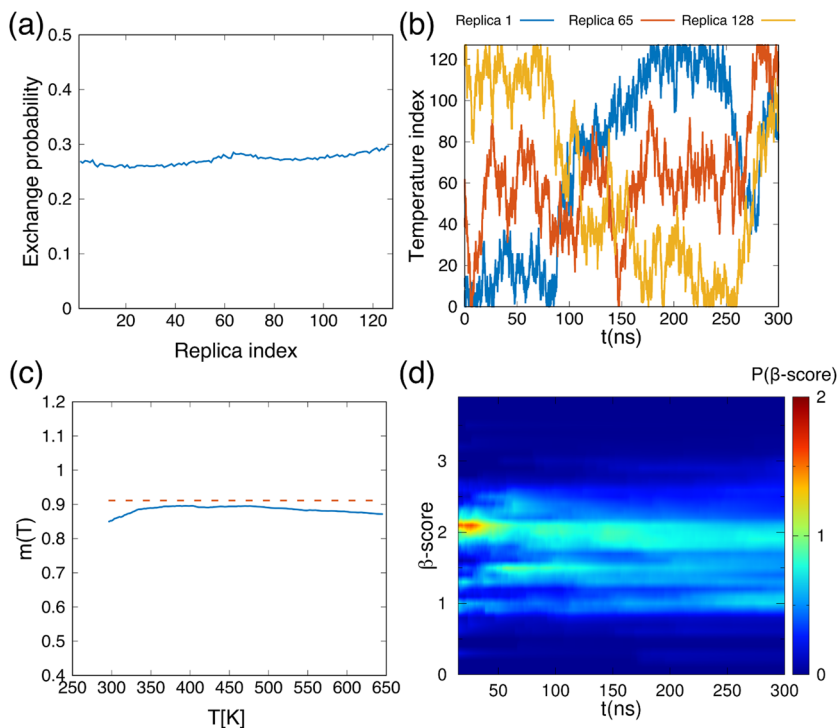


FIG. 2. Convergence test of the simulation results in various criteria. (a) The exchange probability for all replicas. The actual exchange rate is in the range of 0.27-0.3 close to our targeted rate 0.3. (b) Temperatures at time t for a few selected replicas. The replicas presented traverse the wide range of temperatures within 300 ns. (c) Mixing parameters $m(t)$ gauging the quality of mixing. The $m(t)$ is close to the ideal value (dashed red line) for all replicas. (d) The change of distribution of the β -sheet collective variable as a function of time. The distribution shows little change after 100 ns ensuring the convergence of the secondary structure after 100 ns.

$$S = \sum_{\alpha} n[\text{RMSD}(\{\mathbf{R}_i\}_{i \in \Omega_{\alpha}}, \{\mathbf{R}^0\})] \quad (1)$$

$$n(\text{RMSD}) = \frac{1 - (\text{RMSD}/0.1)^8}{1 - (\text{RMSD}/0.1)^{12}} \quad (2)$$

what does bad $m(T)$ look like?

introduced by Pietrucci and Laio,³⁶ which measures the secondary structure propensity in a continuous scale. To test if the secondary structure of our simulated proteins is properly sampled in our simulation protocol, we calculated the distribution of the collective variable for β -sheet structures (or β -score) in the polyQ region, which is known as the slowest varying collective variable. Figure 2(d) illustrates the evolution of the secondary structure propensity as a function of the time period we sampled, showing no significant change after 200 ns. Thus, with these considerations, we concluded that our T-REMD simulations are reasonably converged.

Mean-squared displacement and relaxation time

Next, we compared our results from T-REMD with the ones from unbiased MD to test if the same results can be reproduced within a reasonable amount of time in MD. As previously emphasized, since httex1 is an intrinsically disordered protein (IDP), it is significantly more challenging to obtain a converged result than normal proteins. Hence, the system is relaxed within simulation time only after visiting different thermodynamic states as much as possible throughout the simulations.

The mean-squared displacement (msd), R^2 , of individual particles is a straightforward and rigorous measure to gauge such relaxation. However, the motions of individual atoms are coupled with the collective motion of all atoms, i.e., the translation and the rotation of the entire protein, and thus these collective motions need to be removed beforehand. To do so, we performed pair-wise alignment for two conformations at distant time and then calculated the msd for C_{α} atoms between them [Fig. 3(a)].

Since glutamines have similar chemical structures with backbones, a relatively long polyQ tract has very strong

intra-hydrogen bonding propensity, which makes glutamines in a polyQ chain “sticky.”¹⁹ Thus, the dynamics of the polyQ region are expected to be much slower than other domains, and consequently the relaxation of the whole system will be mostly determined by the relaxation of the polyQ region. For this reason, we analyzed the relaxation of the polyQ region only.

The msd (R^2) for the polyQ region of Q56 at three selected temperatures (298 K, 420 K, and 600 K) are presented in Fig. 3(b). If we assume that the polyQ tract is completely disordered so that it can be modeled as an ideal polymer, then the mean squared distance between C_{α} atoms will be $R_0^2 = \frac{1}{2N^2} \sum_{i,j} (r_i - r_j)^2 = \frac{(N-1)a^2}{6}$, where N is the number of the monomers in the polymer and a is the monomer-monomer distance.³⁷ Plugging $N = 55$ and $a = 0.38$ nm into the above equation, then we obtain $R_0^2 = 1.3$ nm². Indeed, we observed in Fig. 3(b) that values of R^2 (for 600 K data) start to deviate from the initial linear relation roughly around 1.3 nm². Thus, we considered the time when $\langle R^2 \rangle$, which is an averaged R^2 over all trajectories, reaches around 1.3 nm² as the approximate relaxation time for the polyQ.

By contrast, the 420 K and 298 K data which are several orders of magnitude smaller than R_0^2 show the continuously increasing trend of R^2 at 100 ns, indicating that the overall httex1 structure was not properly relaxed during the simulation time. More importantly, the R^2 from five trajectories seems to be homogeneous at 600 K but quite heterogeneous at 420 K and 298 K. Such slow and heterogeneous dynamics are a signature of a glassy system.^{38–41}

Since our unbiased MD simulation time was much shorter than the expected relaxation time, we use extrapolation to estimate the relaxation time at lower temperatures. At all temperatures, our data unanimously show the power-law relation at small R^2 values. Thus, with the scaling relation $\langle R^2 \rangle \propto \Delta t^{\xi}$,

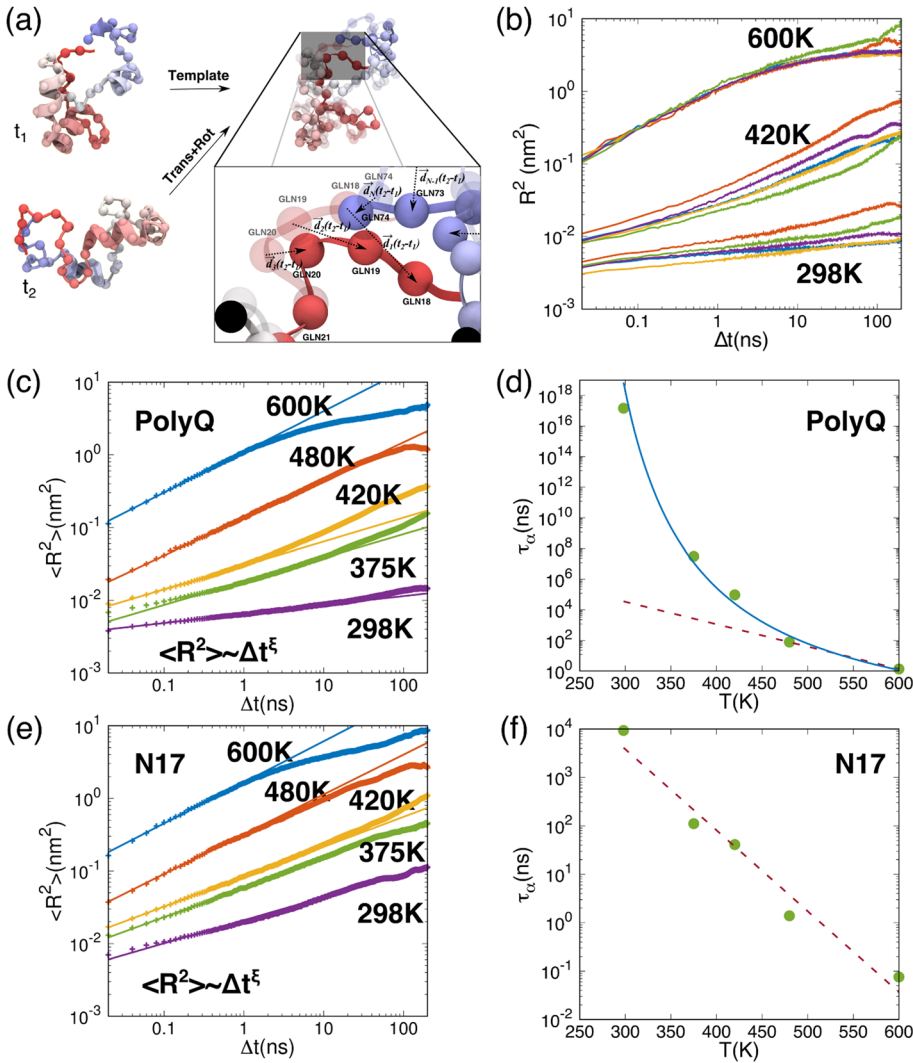


FIG. 3. Relaxation of the system measured by mean-squared displacements (R^2). (a) The schematic pictures explaining the algorithm. For the calculation of the R^2 between time t_1 (top left) and t_2 (bottom left), the conformation at t_2 was aligned with t_1 to remove the collective motions of the protein (translation and rotation). Then, the squared displacement, d^2 , of each C_α atom between two conformations was calculated for aligned structures. R^2 is defined as an average d^2 over all C_α atoms (right). (b) The estimated R^2 of the polyQ region of Q56 at 298 K, 375 K, 420 K, and 600 K from five independent unbiased simulations for each temperature. Whereas 600 K data show homogeneous and saturated R^2 at ~ 10 nm² regardless of trajectories, the R^2 for 420 K and 298 K varies with the trajectory number or heterogeneous and increases at 300 ns exemplifying dynamic heterogeneity of low temperature data. (c) The average R^2 and their power-law fits to $\langle R^2 \rangle \propto t^\xi$, where $\langle R^2 \rangle$ is an average msd, t is the time, and ξ is a scaling exponent. (d) The fitting of the relaxation time τ to the Vogel-Fulcher-Tammann (VFT) law. τ was estimated by solving $\langle R^2 \rangle_0 = \eta \tau^\xi$, with $\langle R^2 \rangle_0 = 1.3$ nm². τ was fit well to VFT law with $T_0 = 220$ K and $B = 4218$ while deviating from Arrhenius's equation drawn in a red dashed line. (e) The average msd and (f) the relaxation time for N17. In contrast to polyQ, the relaxation time fits well to the exponential, indicating a liquid-like state.

where d is the msd, Δt is time interval, and ξ is the exponent of the power-law fit, our fit using the msd data from short time scale ($\Delta t < 2$ ns) shows fairly good agreements with the data [Fig. 3(c), with an extra data set for $T = 480$ K]. Next, the relaxation time was obtained by solving $\langle R^2 \rangle = \eta \tau^\xi$ or $\tau = (\langle R^2 \rangle / \eta)^{1/\xi}$, where η is a coefficient. We assumed that the structure is relaxed when the average msd, $\langle R^2 \rangle$, reaches 1.3 nm². The estimated relaxation time τ_α for each temperature is presented in Fig. 3(d).

Strikingly, the estimated relaxation time, τ_α , exhibits non-Arrhenius behavior, which is described by the deviation from the exponential fits [dashed red line in Fig. 3(d)]. Instead, τ_α is fit well to Vogel-Fulcher-Tammann (VFT) law, defined as

$$\tau = \tau_0 \exp\left(\frac{B}{T - T_0}\right),$$

where τ_0 is the microscopic relaxation time, B is the VFT coefficient, and T_0 is an ideal glass transition temperature (where the true glass transition occurs). The good agreement of the VFT fitting to the data indicates that the dynamics of the polyQ region become extremely sluggish as the temperature is lowered. As it approaches to a critical temperature $T_0 = 229$ K, the dynamic behavior of the polyQ region diverges

like glass-forming liquids characterized by overwhelmingly slow dynamics and the existence of an exponentially large number of metastable states.⁴² This observation is consistent with the glassy behavior of a pure polyQ chain without N17 and C38.⁴³ Due to such a divergently slow dynamics, τ_α becomes an order of hundred microseconds at 420 K and several years at 298 K, which is not easily accessible even with the most advanced computing facilities.

We compared the msd of the polyQ regions with that of N17 which is known to be more flexible as well as unstructured^{17–19} [Figs. 3(e) and 3(f)]. Since N17 has only 17 amino acids, for a fair comparison, we defined the relaxation time, τ_α , as the time when the msd reaches $R_g^2 = \frac{(N-1)a^2}{6} = \frac{16 \times 0.38^2}{6} = 0.39$ nm² instead of 1.3 nm². The estimated τ_α for N17 fits well to the exponential function $\tau_\alpha = \tau_0 \exp(AT)$ [red dashed line in Fig. 3(f)], implying that N17 dynamics are governed by Arrhenius's equation, and thus remains in disordered liquid-like states than the glassy ones. These observations confirm that the divergently sluggish dynamics upon cooling seem to be specific to the highly viscous polyQ region.

We also compared the msd of unbiased MD with T-REMD data to see how the enhanced sampling can facilitate

the relaxation of the system. Figure S2 of the [supplementary material](#) shows the msd averaged over the 128 replicas reconstituted along replica indices. Interestingly, the msd of T-REMD simulation lies between the ones obtained from unbiased MD simulations at 480 K and 600 K, indicating the significant acceleration of the dynamics for both polyQ and N17 regions.

Secondary structure relaxation

The relaxation of secondary structures was examined as follows. First, we assumed that the secondary structure derived from our T-REMD data is fully converged so that its distribution can represent the “true” thermodynamic ensemble of structures as benchmark. Then, the secondary structures in the unbiased MD simulations will be relaxed as the time evolves and finally reproduce the same distribution as that of T-REMD as $t \rightarrow \infty$. Based on these assumptions, we conjectured that the relaxation of secondary structures is assessed by the similarity of the secondary structure distribution between the samples collected from the two simulation methods.

For simplicity, we only analyzed α -helix and β -sheet structures. The two-dimensional distribution of the α -helix and β -sheet collective variables (or scores) is computed for both T-REMD with entire simulation time and the unbiased MD within a period of 300 ns (top panels of Fig. 4). Then, we

defined the region where the potential of mean-force (PMF) of the secondary structure scores is within the 2kT window from the lowest (referred as “PMF-2kT” in the following) for both T-REMD and MD. The comparison between PMF-2kT of T-REMD and MD is shown in the middle panel of Fig. 4, suggesting that the result from MD approaches, but not anywhere near the full convergence to that from T-REMD. Based on this, we define the ratio of the overlap area to the total area of the PMF-2kT for T-REMD as the extent of secondary structure relaxation (bottom panel of Fig. 4).

The estimated ratios of secondary structure relaxation parameters are presented in Fig. 5(a). To quantify the secondary structure relaxation, we utilized the power-law fitting and the extrapolation method similar to the msd analyses. The secondary structure relaxation time, τ_α , was defined as the time when the secondary structure relaxation parameter reaches 0.8; in other words, the PMF-2kT of the unbiased MD can cover 80% of the PMF-2kT of T-REMD. Strikingly, the relaxation time estimated from the power-law fittings also exhibits non-Arrhenius behavior as the temperature is lowered [Fig. 5(b)]. Furthermore, the fitting parameter $T_0 = 223$ K, i.e., the relaxation time diverges at 223 K, which is close to the transition temperature, $T_0 = 229$ K, obtained from the msd analysis. These results again reassure that the long polyQ chain behaves like a glass-forming material above its critical temperature.

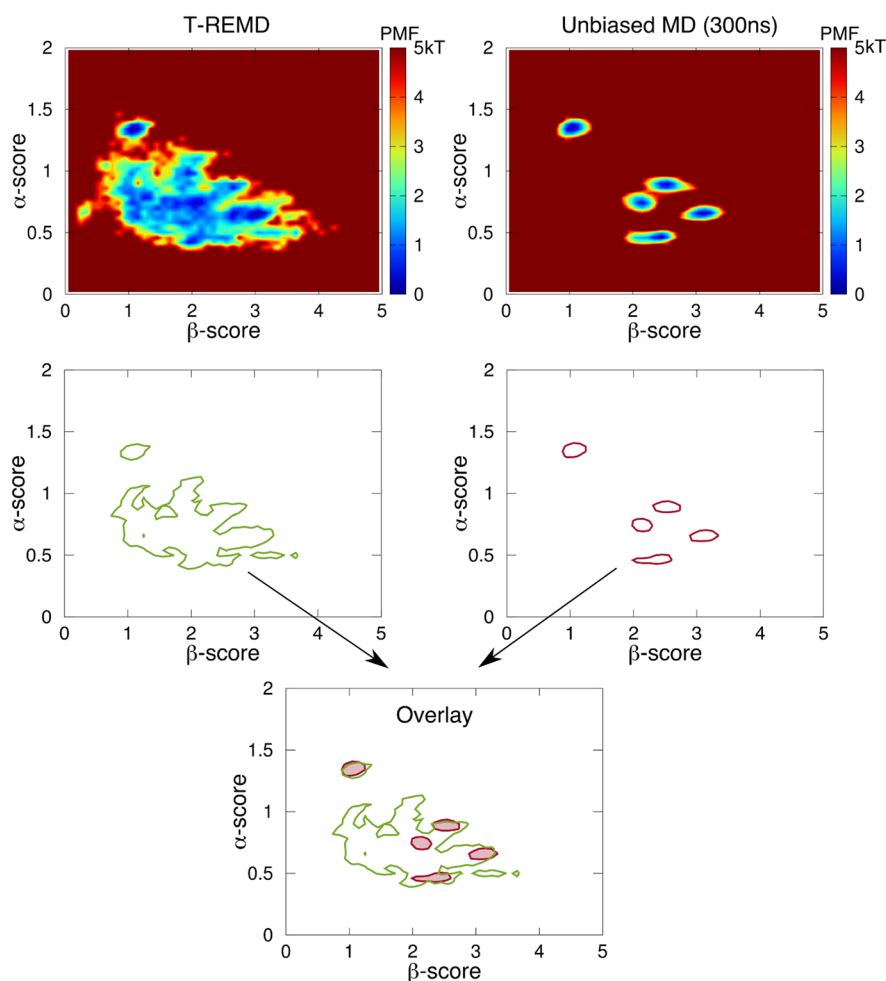


FIG. 4. Illustration of algorithm to calculate secondary structure relaxation. The 2D potential of mean force (PMF) for α -helix and β -sheet collective variables is calculated for thermodynamic ensembles (T-REMD, top left) and collective conformations from unbiased MD within time span t (top right). The areas with 2kT energy difference from the lowest PMF, PMF-2kT, are computed for both (middle panels). The overlap between PMF-2kT for the T-REMD and the targeted samples is estimated (bottom). Finally, the ratio of the overlap to the total area of the PMF-2kT for the T-REMD is defined as a measure of the secondary structure relaxation.

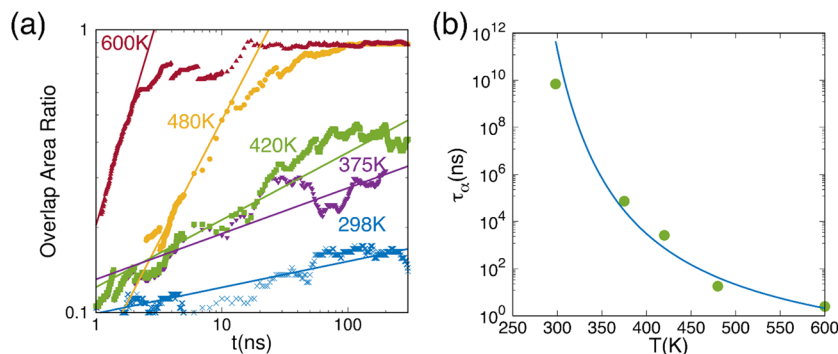


FIG. 5. (a) The secondary structure relaxation parameters for the polyQ of Q56 at 298 K, 375 K, 420 K, 480 K, and 600 K. Lines indicate their power-law fits. (b) The relaxation time (green circles) estimated similarly to the msd and its fitting results to VFT law (blue line). Surprisingly, the secondary structure relaxation time seems to diverge at $T_0 = 224$ K close to $T_0 = 220$ K for the msd.

Cis-trans isomerization

Each httex1 contains a significant number of proline (Pro) residues (28 in total). It is known that unlike other amino acids that predominantly favor *trans*-peptide bonds, a Pro residue usually has a non-negligible propensity to *cis*-peptide bonds. The fraction of *cis*-Pro-peptide bonds can vary according to the types of amino acids preceding the Pro residue and generally ranges from 3% to 10% for amino acids other than the glycine and aromatic ones. Since the kinetics of *cis-trans* isomerization is several orders of magnitudes slower than any other protein folding processes, it is often considered as a key determinant of the folding kinetics. Thus, whether such a *cis-trans* isomerization can be observed through simulations or not is another assessment of proper relaxation of httex1.

From T-REMD trajectories of Q56, we analyzed the representative structures that were selected from the cluster analysis based on the root-mean-squared displacement of C_α atoms and GROMOS algorithm with 3.5 Å cutoff. Surprisingly, *cis*-peptide bonds were found in C38 regions

even at 298 K. Figure 6(a) illustrates the existence of *cis*-peptide bonds in C38. In particular, it was relatively abundant in a flexible “hinge” region (residue number 85-100 in Q56) composed of leucines and glutamines as well as prolines. Figure 6(b) shows the fraction of the *cis*-bond for each residue with the maximum fraction of 2.3% at Leu86-Pro87. One may argue that this amount of the *cis*-peptide bond is too small to be significant. However, first, we cannot completely rule out the possibility that such a small fraction of conformation might affect its binding with other important proteins (as HTT has a very broad protein-protein interaction network²) and initiate some pathogenic pathways. Second, because httex1 is structurally heterogeneous, there is no dominant structure unlike general globular proteins (Fig. S1 of the [supplementary material](#)). As a result, the significance of each structural cluster is non-negligible even if its population is relatively small. Indeed, the conformation with the *cis*-peptide bond in Leu86-Pro87 was identified as the representative structure of the 7th most populated clusters having 3.2% population of total conformations [Fig. 6(c)]. By contrast, in unbiased MD, the *cis*-peptide bonds were not observed

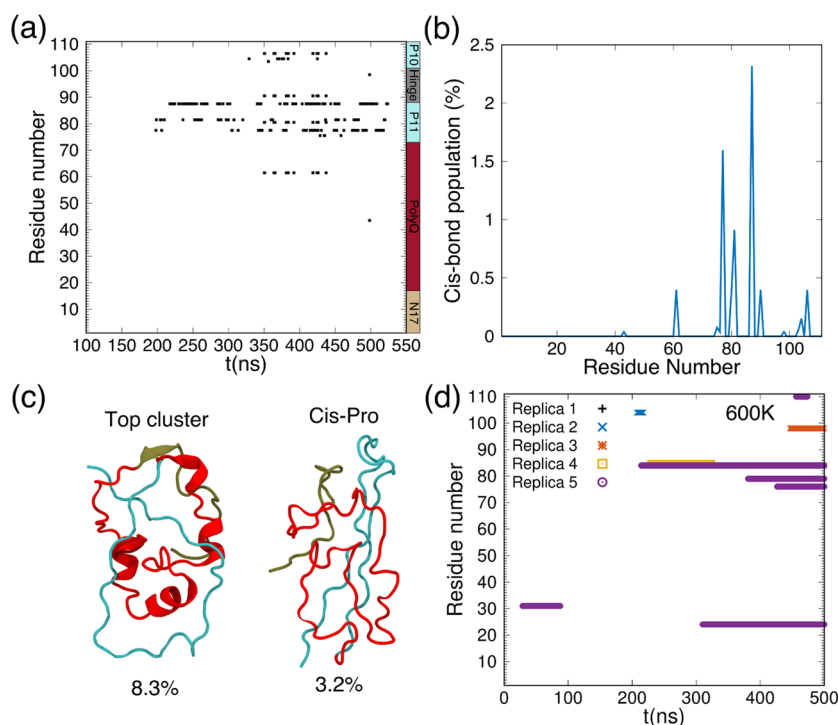


FIG. 6. (a) *Cis*-peptide bonds (black squares) found in T-REMD of Q56 at 298 K. Most *cis*-bonds are located in C38, in particular, a relatively flexible hinge region. (b) Statistics in *cis*-bonds by the residue number. The maximum percentage of *cis*-bonds is below 2.5%. (c) The representative structures of the most populated cluster (left) and the 7th most populated one containing the *cis*-bonds (right). (d) The *cis*-bonds in unbiased MD at 600 K. Other temperatures (298 K, 420 K, and 480 K) did not show any such *trans-cis* or *cis-trans* transition at all.

until the temperature was raised up to 600 K [within 500 ns; Fig. 6(d)].

DISCUSSION

In this paper, we examine the dynamics of the httex1 protein by analyzing T-REMD and unbiased MD simulation results to highlight the advantages of enhanced sampling in the study of intrinsically disordered proteins. Surprisingly, our analyses reveal that the relaxation time of the polyQ region diverges upon cooling following the VFT law as expected for glass-forming liquids. The estimated relaxation time is several hundred microseconds even at 420 K and several years at 298 K, both of which are not easily accessible in normal MD simulation even with cutting-edge computing facilities. The secondary structure analysis of the polyQ region also confirms non-Arrhenius behavior accompanied by diverging relaxation time with decreasing temperature. We speculate that such sluggish dynamics of the polyQ region are due to a very strong intra-hydrogen-bond propensity that makes glutamine residues “viscous” or “sticky.”¹⁹ Hence, this kind of glassy behavior is possibly applicable to not only httex1 but also general polyQ diseases related proteins such as spinocerebellar ataxias (SCA) and pallidolusian atrophy (DRPLA). However, without experimental evidences so far, such an expectation might be very speculative.

These observations pose another question: what is the implication of the glassy behavior of the polyQ region in the disease mechanism? We argue that such sluggish dynamics should be understood by making an analogy between the httex1 and the super-cooled liquid. The thermodynamic states of the super-cooled liquid near the critical temperature are characterized by an exponentially large number of metastable states separated by large energy barriers. This is different from the liquid upon annealing that approaches to a single global minimum or a sub-exponential number of local minima corresponding to a crystal or inherent structures,⁴⁴ respectively. For the protein, we can make a similar argument. General globular proteins have a single or a few native conformations, and the system will fall into the native state regardless of initial conformations. There might exist metastable states, but the barrier is not high enough so that it can be overcome by thermal fluctuation within observable time scale. However, the httex1 or other polyQ proteins have very strong internal interactions yielding metastable states that are mutually exclusive within an observable time scale at room temperature. Therefore, the ensemble of httex1 exon-1 might be structurally heterogeneous rather than homogeneous even after the system is equilibrated for long time. We emphasize that the glassy dynamics might not be universal for general IDPs but unique for the polyQ proteins. For instance, the lower bound of the folding-unfolding rate constants for the ABP1 SH3 domain having 58 residues is 17.1 s^{-1} , which is much shorter than the relaxation time of the httex1.⁴⁵

Another remark is the necessity of high temperature simulations. One might expect that 400 K is high enough for modern force fields to relax the system, fully unfold the protein, and sample accurate thermodynamic ensemble-structures because it is much higher than the normal melting temperatures of

general proteins. However, our simulations show that for long polyQ chains, the relaxation time at 420 K is on the order of hundreds microseconds which is much longer than time scales usually used in all-atom simulations. Furthermore, the *cis-trans* transition of peptide bonds occurs only at or above 530 K so that the *cis*-peptide bond conformation might not be sampled within the observable time scale without high temperature above 500 K even in T-REMD. These observations are consistent with the simulation results by García and his colleague⁴⁶ using the same OPLS force field. Generally, the *trans*-to-*cis* isomerization occurring at 500 K or higher temperature is slightly force-field dependent.⁴⁶ As speculated above, these *cis*-peptide bond conformations might be functionally important.

Kinetics of the *cis-trans* transition is another interesting topic. It would be desirable to calculate the rate constant of the transition from our simulations. Because the neurodegeneration due to the HTT aggregation takes up to several decades to be developed, very slow processes such as the *cis-trans* transition might be crucial for HD progression. Thus, it would be advantageous if we can figure out the rate of the transition and compare it with the HD development time table. Indeed, advanced algorithms to estimate the rate constants from T-REMD or enhanced sampling methods have been proposed.^{47–49} However, we were not able to have reliable rate constants for the transition due to the paucity of the data from unbiased MD simulations. Therefore, the kinetic analysis with much longer simulation trajectories might be desirable for further studies.

SUPPLEMENTARY MATERIAL

See [supplementary material](#) for the details of simulation temperatures in the T-REMD method, representative structures of Q56 at room temperature, comparison of the msd for T-REMD and unbiased MD, and the time-series of secondary structures.

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